

ROOT FINDING

To this point you have found solutions to equations almost exclusively by algebraic manipulation. This is possible only for the artificially simple equations of problem sets and tests. In the “real world” it is very common to encounter equations that cannot be solved by algebraic manipulation. For example, you found, by completing a square, that the solutions to the quadratic equation $ax^2 + bx + c = 0$ are $x = (-b \pm \sqrt{b^2 - 4ac})/2a$. But it is known that there simply does not exist a corresponding formula for the roots of a general polynomial of degree five or more. Fortunately, encountering such an equation is not the end of the world, because usually one does not need to know the solutions exactly. One only needs to know them to within some specified degree of accuracy. For example, one rarely needs to know π to more than a few decimal places. There is a whole subject, called numerical analysis, that concerns using algorithms to solve equations (and perform other tasks) approximately, to any desired degree of accuracy.

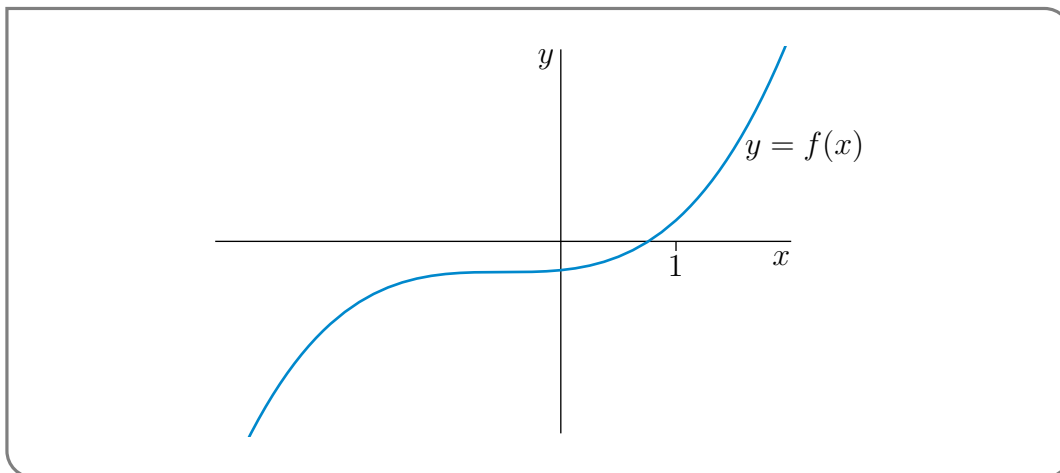
We have already had, in Examples [1.6.14](#) and [1.6.15](#), and the lead up to them, a really quick introduction to the bisection method, which is a crude, but effective, algorithm for finding approximate solutions to equations of the form $f(x) = 0$. We shall shortly use a little calculus to derive a very efficient algorithm for finding approximate solutions to such equations. But first here is a simple example which provides a review of some of the basic ideas of root finding and the bisection method.

Example C.0.1

Suppose that we are given some function $f(x)$ and we have to find solutions to the equation $f(x) = 0$. To be concrete, suppose that $f(x) = 8x^3 + 12x^2 + 6x - 15$. How do we go about solving $f(x) = 0$? To get a rough idea of the lay of the land, sketch the graph of $f(x)$. First observe that

- when x is very large and negative, $f(x)$ is very large and negative
- when x is very large and positive, $f(x)$ is very large and positive
- when $x = 0$, $f(x) = f(0) = -15 < 0$
- when $x = 1$, $f(x) = f(1) = 11 > 0$
- $f'(x) = 24x^2 + 24x + 6 = 24(x^2 + x + \frac{1}{4}) = 24(x + \frac{1}{2})^2 \geq 0$ for all x . So $f(x)$ increases monotonically with x . The graph has a tangent of slope 0 at $x = -\frac{1}{2}$ and tangents of strictly positive slope everywhere else.

This tells us that the graph of $f(x)$ looks like



Since $f(x)$ strictly increases¹ as x increases, $f(x)$ can take the value zero for at most one value of x .

- Since $f(0) < 0$ and $f(1) > 0$ and f is continuous, $f(x)$ must pass through 0 as x travels from $x = 0$ to $x = 1$, by Theorem 1.6.12 (the intermediate value theorem). So $f(x)$ takes the value zero for some x between 0 and 1. We will often write this as “the root is $x = 0.5 \pm 0.5$ ” to indicate the uncertainty.
- To get closer to the root, we evaluate $f(x)$ halfway between 0 and 1.

$$f\left(\frac{1}{2}\right) = 8\left(\frac{1}{2}\right)^3 + 12\left(\frac{1}{2}\right)^2 + 6\left(\frac{1}{2}\right) - 15 = -8$$

Since $f\left(\frac{1}{2}\right) < 0$ and $f(1) > 0$ and f is continuous, $f(x)$ must take the value zero for some x between $\frac{1}{2}$ and 1. The root is 0.75 ± 0.25 .

- To get still closer to the root, we evaluate $f(x)$ halfway between $\frac{1}{2}$ and 1.

$$f\left(\frac{3}{4}\right) = 8\left(\frac{3}{4}\right)^3 + 12\left(\frac{3}{4}\right)^2 + 6\left(\frac{3}{4}\right) - 15 = -\frac{3}{8}$$

Since $f\left(\frac{3}{4}\right) < 0$ and $f(1) > 0$ and f is continuous, $f(x)$ must take the value zero for some x between $\frac{3}{4}$ and 1. The root is 0.875 ± 0.125 .

- And so on.

Example C.0.1

The root finding strategy used in Example C.0.1 is called the bisection method. The bisection method will home in on a root of the function $f(x)$ whenever

- $f(x)$ is continuous ($f(x)$ need not have a derivative) and

1 By “ $f(x)$ is strictly increasing” we mean that $f(a) < f(b)$ whenever $a < b$. As $f'(x) > 0$ for all $x \neq -\frac{1}{2}$, $f(x)$ is strictly increasing even as x passes through $-\frac{1}{2}$. For example, for any $x > -\frac{1}{2}$, the mean value theorem (Theorem 2.13.4), tells us that there is a c strictly between $-\frac{1}{2}$ and x such that $f(x) - f\left(-\frac{1}{2}\right) = f'(c)\left(x + \frac{1}{2}\right) > 0$.

- you can find two numbers $a_1 < b_1$ with $f(a_1)$ and $f(b_1)$ being of opposite sign.

Denote by I_1 the interval $[a_1, b_1] = \{x \mid a_1 \leq x \leq b_1\}$. Once you have found the interval I_1 , the bisection method generates a sequence $I_1, I_2, I_3, \dots$ of intervals by the following rule.

Equation C.0.2 (bisection method).

Denote by $c_n = \frac{a_n + b_n}{2}$ the midpoint of the interval $I_n = [a_n, b_n]$. If $f(c_n)$ has the same sign as $f(a_n)$, then

$$I_{n+1} = [a_{n+1}, b_{n+1}] \quad \text{with} \quad a_{n+1} = c_n, \quad b_{n+1} = b_n$$

and if $f(c_n)$ and $f(a_n)$ have opposite signs, then

$$I_{n+1} = [a_{n+1}, b_{n+1}] \quad \text{with} \quad a_{n+1} = a_n, \quad b_{n+1} = c_n$$

This rule was chosen so that $f(a_n)$ and $f(b_n)$ have opposite sign for every n . Since $f(x)$ is continuous, $f(x)$ has a zero in each interval I_n . Thus each step reduces the error bars by a factor of 2. That isn't too bad, but we can come up with something that is much more efficient. We just need a little calculus.

C.1 ▲ Newton's Method

Newton's method², also known as the Newton-Raphson method, is another technique for generating numerical approximate solutions to equations of the form $f(x) = 0$. For example, one can easily get a good approximation to $\sqrt{2}$ by applying Newton's method to the equation $x^2 - 2 = 0$. This will be done in Example C.1.2, below.

Here is the derivation of Newton's method. We start by simply making a guess for the solution. For example, we could base the guess on a sketch of the graph of $f(x)$. Call the initial guess x_1 . Next recall, from Theorem 2.3.2, that the tangent line to $y = f(x)$ at $x = x_1$ is $y = F(x)$, where

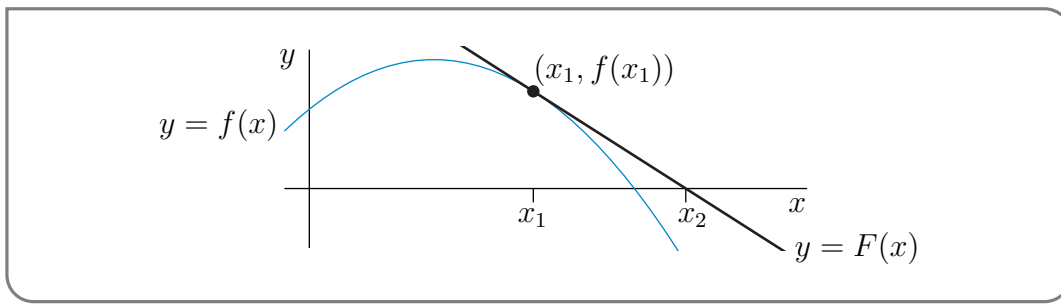
$$F(x) = f(x_1) + f'(x_1)(x - x_1)$$

Usually $F(x)$ is a pretty good approximation to $f(x)$ for x near x_1 . So, instead of trying to solve $f(x) = 0$, we solve the linear equation $F(x) = 0$ and call the solution x_2 .

$$\begin{aligned} 0 = F(x) &= f(x_1) + f'(x_1)(x - x_1) \iff x - x_1 = -\frac{f(x_1)}{f'(x_1)} \\ &\iff x = x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \end{aligned}$$

Note that if $f(x)$ were a linear function, then $F(x)$ would be exactly $f(x)$ and x_2 would solve $f(x) = 0$ exactly.

2 The algorithm that we are about to describe grew out of a method that Newton wrote about in 1669. But the modern method incorporates substantial changes introduced by Raphson in 1690 and Simpson in 1740.



Now we repeat, but starting with the (second) guess x_2 rather than x_1 . This gives the (third) guess $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$. And so on. By way of summary, Newton's method is

1. Make a preliminary guess x_1 .
2. Define $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$.
3. Iterate. That is, for each natural number n , once you have computed x_n , define

Equation C.1.1 (Newton's method).

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Example C.1.2 (Approximating $\sqrt{2}$)

In this example we compute, approximately, the square root of two. We will of course pretend that we do not already know that $\sqrt{2} = 1.41421 \dots$. So we cannot find it by solving, approximately, the equation $f(x) = x - \sqrt{2} = 0$. Instead we apply Newton's method to the equation

$$f(x) = x^2 - 2 = 0$$

Since $f'(x) = 2x$, Newton's method says that we should generate approximate solutions by iteratively applying

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n}$$

We need a starting point. Since $1^2 = 1 < 2$ and $2^2 = 4 > 2$, the square root of two must be between 1 and 2, so let's start Newton's method with the initial guess $x_1 = 1.5$. Here goes³:

$$\begin{aligned} x_1 &= 1.5 \\ x_2 &= \frac{1}{2}x_1 + \frac{1}{x_1} = \frac{1}{2}(1.5) + \frac{1}{1.5} \\ &= 1.416666667 \end{aligned}$$

³ The following computations have been carried out in double precision, which is computer speak for about 15 significant digits. We are displaying each x_n rounded to 10 significant digits (9 decimal places). So each displayed x_n has not been impacted by roundoff error, and still contains more decimal places than are usually needed.

$$\begin{aligned}
 x_3 &= \frac{1}{2}x_2 + \frac{1}{x_2} = \frac{1}{2}(1.416666667) + \frac{1}{1.416666667} \\
 &= 1.414215686 \\
 x_4 &= \frac{1}{2}x_3 + \frac{1}{x_3} = \frac{1}{2}(1.414215686) + \frac{1}{1.414215686} \\
 &= 1.414213562 \\
 x_5 &= \frac{1}{2}x_4 + \frac{1}{x_4} = \frac{1}{2}(1.414213562) + \frac{1}{1.414213562} \\
 &= 1.414213562
 \end{aligned}$$

It looks like the x_n 's, rounded to nine decimal places, have stabilized to 1.414213562. So it is reasonable to guess that $\sqrt{2}$, rounded to nine decimal places, is exactly 1.414213562. Recalling that all numbers $1.4142135615 \leq y < 1.4142135625$ round to 1.414213562, we can check our guess by evaluating $f(1.4142135615)$ and $f(1.4142135625)$. Since $f(1.4142135615) = -2.5 \times 10^{-9} < 0$ and $f(1.4142135625) = 3.6 \times 10^{-10} > 0$ the square root of two must indeed be between 1.4142135615 and 1.4142135625.

Example C.1.2

Example C.1.3 (Approximating π)

In this example we compute, approximately, π by applying Newton's method to the equation

$$f(x) = \sin x = 0$$

starting with $x_1 = 3$. Since $f'(x) = \cos x$, Newton's method says that we should generate approximate solutions by iteratively applying

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{\sin x_n}{\cos x_n} = x_n - \tan x_n$$

Here goes

$$\begin{aligned}
 x_1 &= 3 \\
 x_2 &= x_1 - \tan x_1 = 3 - \tan 3 \\
 &= 3.142546543 \\
 x_3 &= 3.142546543 - \tan 3.142546543 \\
 &= 3.141592653 \\
 x_4 &= 3.141592653 - \tan 3.141592653 \\
 &= 3.141592654 \\
 x_5 &= 3.141592654 - \tan 3.141592654 \\
 &= 3.141592654
 \end{aligned}$$

Since $f(3.1415926535) = 9.0 \times 10^{-11} > 0$ and $f(3.1415926545) = -9.1 \times 10^{-11} < 0$, π must be between 3.1415926535 and 3.1415926545. Of course to compute π in this way, we (or at least our computers) have to be able to evaluate $\tan x$ for various values of x . Taylor expansions can help us do that. See Example [3.4.22](#).

Example C.1.3

Example C.1.4 (wild instability)

This example illustrates how Newton's method can go badly wrong if your initial guess is not good enough. We'll try to solve the equation

$$f(x) = \arctan x = 0$$

starting with $x_1 = 1.5$. (Of course the solution to $f(x) = 0$ is just $x = 0$; we chose $x_1 = 1.5$ for demonstration purposes.) Since the derivative $f'(x) = \frac{1}{1+x^2}$, Newton's method gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - (1 + x_n^2) \arctan x_n$$

So⁴

$$x_1 = 1.5$$

$$x_2 = 1.5 - (1 + 1.5^2) \arctan 1.5 = -1.69$$

$$x_3 = -1.69 - (1 + 1.69^2) \arctan(-1.69) = 2.32$$

$$x_4 = 2.32 - (1 + 2.32^2) \arctan(2.32) = -5.11$$

$$x_5 = -5.11 - (1 + 5.11^2) \arctan(-5.11) = 32.3$$

$$x_6 = 32.3 - (1 + 32.3^2) \arctan(32.3) = -1575$$

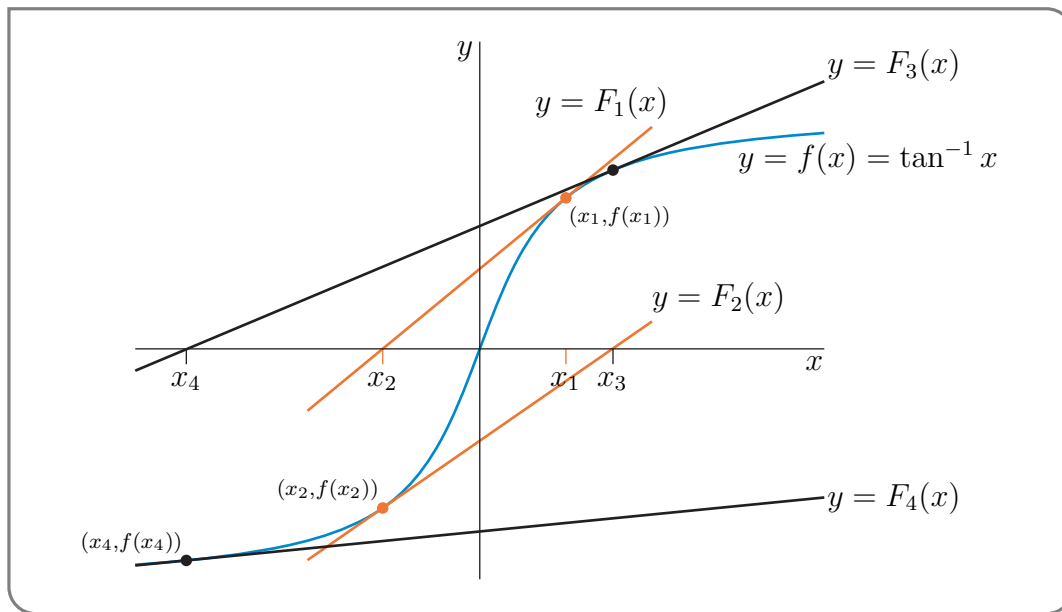
$$x_7 = 3,894,976$$

Looks pretty bad! Our x_n 's are not settling down at all!

The figure below shows what went wrong. In this figure, $y = F_1(x)$ is the tangent line to $y = \arctan x$ at $x = x_1$. Under Newton's method, this tangent line crosses the x -axis at $x = x_2$. Then $y = F_2(x)$ is the tangent to $y = \arctan x$ at $x = x_2$. Under Newton's method, this tangent line crosses the x -axis at $x = x_3$. And so on.

The problem arose because the x_n 's were far enough from the solution, $x = 0$, that the tangent line approximations, while good approximations to $f(x)$ for $x \approx x_n$, were very

4 Once again, the following computations have been carried out in double precision. This time, it is clear that the x_n 's are growing madly as n increases. So there is not much point to displaying many decimal places and we have not done so.



poor approximations to $f(x)$ for $x \approx 0$. In particular, $y = F_1(x)$ (i.e. the tangent line at $x = x_1$) was a bad enough approximation to $y = \arctan x$ for $x \approx 0$ that $x = x_2$ (i.e. the value of x where $y = F_1(x)$ crosses the x -axis) is farther from the solution $x = 0$ than our original guess $x = x_1$. If we had started with $x_1 = 0.5$ instead of $x_1 = 1.5$, Newton's method would have succeeded very nicely:

$$x_1 = 0.5 \quad x_2 = -0.0796 \quad x_3 = 0.000335 \quad x_4 = -2.51 \times 10^{-11}$$

Example C.1.4

Example C.1.5 (interest rate)

A car dealer sells a new car for \$23,520. He also offers to finance the same car for payments of \$420 per month for five years. What interest rate is this dealer charging?

Solution. By way of preparation, we'll start with a simpler problem. Suppose that you will have to make a single \$420 payment n months in the future. The simpler problem is to determine how much money you have to deposit now in an account that pays an interest rate of $100r\%$ per month, compounded monthly⁵, in order to be able to make the \$420 payment in n months.

Let's denote by P the initial deposit. Because the interest rate is $100r\%$ per month, compounded monthly,

- the first month's interest is $P \times r$. So at the end of month #1, the account balance is $P + Pr = P(1 + r)$.
- The second month's interest is $[P(1 + r)] \times r$. So at the end of month #2, the account balance is $P(1 + r) + P(1 + r)r = P(1 + r)^2$.
- And so on.

5 "Compounded monthly", means that, each month, interest is paid on the accumulated interest that was paid in all previous months.

- So at the end of n months, the account balance is $P(1+r)^n$.

In order for the balance at the end of n months, $P(1+r)^n$, to be \$420, the initial deposit has to be $P = 420(1+r)^{-n}$. That is what is meant by the statement “The present value⁶ of a \$420 payment made n months in the future, when the interest rate is $100r\%$ per month, compounded monthly, is $420(1+r)^{-n}$.”

Now back to the original problem. We will be making 60 monthly payments of \$420. The present value of all 60 payments is⁷

$$\begin{aligned} 420(1+r)^{-1} + 420(1+r)^{-2} + \cdots + 420(1+r)^{-60} &= 420 \frac{(1+r)^{-1} - (1+r)^{-61}}{1 - (1+r)^{-1}} \\ &= 420 \frac{1 - (1+r)^{-60}}{(1+r) - 1} = 420 \frac{1 - (1+r)^{-60}}{r} \end{aligned}$$

The interest rate $100r\%$ being charged by the car dealer is such that the present value of 60 monthly payments of \$420 is \$23520. That is, the monthly interest rate being charged by the car dealer is the solution of

$$\begin{aligned} 23520 &= 420 \frac{1 - (1+r)^{-60}}{r} & \text{or} & \quad 56 = \frac{1 - (1+r)^{-60}}{r} \\ & & \text{or} & \quad 56r = 1 - (1+r)^{-60} \\ & & \text{or} & \quad 56r(1+r)^{60} = (1+r)^{60} - 1 \\ & & \text{or} & \quad (1 - 56r)(1+r)^{60} = 1 \end{aligned}$$

Set $f(r) = (1 - 56r)(1+r)^{60} - 1$. Then

$$f'(r) = -56(1+r)^{60} + 60(1-56r)(1+r)^{59}$$

or

$$f'(r) = [-56(1+r) + 60(1-56r)](1+r)^{59} = (4 - 3416r)(1+r)^{59}$$

Apply Newton's method with an initial guess of $r_1 = .002$. (That's 0.2% per month or 2.4% per year.) Then

$$\begin{aligned} r_2 &= r_1 - \frac{(1 - 56r_1)(1+r_1)^{60} - 1}{(4 - 3416r_1)(1+r_1)^{59}} = 0.002344 \\ r_3 &= r_2 - \frac{(1 - 56r_2)(1+r_2)^{60} - 1}{(4 - 3416r_2)(1+r_2)^{59}} = 0.002292 \\ r_4 &= r_3 - \frac{(1 - 56r_3)(1+r_3)^{60} - 1}{(4 - 3416r_3)(1+r_3)^{59}} = 0.002290 \\ r_5 &= r_4 - \frac{(1 - 56r_4)(1+r_4)^{60} - 1}{(4 - 3416r_4)(1+r_4)^{59}} = 0.002290 \end{aligned}$$

6 Inflation means that prices of goods (typically) increase with time, and hence \$100 now is worth more than \$100 in 10 years time. The term “present value” is widely used in economics and finance to mean “the current amount of money that will have a specified value at a specified time in the future”. It takes inflation into account. If the money is invested, it takes into account the rate of return of the investment. We recommend that the interested reader do some search-engining to find out more.

7 Don't worry if you don't know how to evaluate such sums. They are called geometric sums, and will be covered in the CLP-2 text. (See (1.1.3) in the CLP-2 text.) In any event, you can check that this is correct, by multiplying the whole equation by $1 - (1+r)^{-1}$. When you simplify the left hand side, you should get the right hand side.

So the interest rate is 0.229% per month or 2.75% per year.

Example C.1.5

C.2 ▲ The Error Behaviour of Newton's Method

Newton's method usually works spectacularly well, provided your initial guess is reasonably close to a solution of $f(x) = 0$. A good way to select this initial guess is to sketch the graph of $y = f(x)$. We now explain why "Newton's method usually works spectacularly well, provided your initial guess is reasonably close to a solution of $f(x) = 0$ ".

Let r be any solution of $f(x) = 0$. Then $f(r) = 0$. Suppose that we have already computed x_n . The error in x_n is $|x_n - r|$. We now derive a formula that relates the error after the next step, $|x_{n+1} - r|$, to $|x_n - r|$. We have seen in (3.4.32) that

$$f(x) = f(x_n) + f'(x_n)(x - x_n) + \frac{1}{2}f''(c)(x - x_n)^2$$

for some c between x_n and x . In particular, choosing $x = r$,

$$0 = f(r) = f(x_n) + f'(x_n)(r - x_n) + \frac{1}{2}f''(c)(r - x_n)^2 \quad (\text{E1})$$

Recall that x_{n+1} is the solution of $0 = f(x_n) + f'(x_n)(x - x_n)$. So

$$0 = f(x_n) + f'(x_n)(x_{n+1} - x_n) \quad (\text{E2})$$

We need to get an expression for $x_{n+1} - r$. Subtracting (E2) from (E1) gives

$$\begin{aligned} 0 = f'(x_n)(r - x_{n+1}) + \frac{1}{2}f''(c)(r - x_n)^2 &\implies x_{n+1} - r = \frac{f''(c)}{2f'(x_n)}(x_n - r)^2 \\ &\implies |x_{n+1} - r| = \frac{|f''(c)|}{2|f'(x_n)|}|x_n - r|^2 \end{aligned}$$

If the guess x_n is close to r , then c , which must be between x_n and r , is also close to r and we will have $f''(c) \approx f''(r)$ and $f'(x_n) \approx f'(r)$ and

$$|x_{n+1} - r| \approx \frac{|f''(r)|}{2|f'(r)|}|x_n - r|^2 \quad (\text{E3})$$

Even when x_n is not close to r , if we know that there are two numbers $L, M > 0$ such that f obeys:

$$(\text{H1}) \quad |f'(x_n)| \geq L$$

$$(\text{H2}) \quad |f''(c)| \leq M$$

(we'll see examples of this below) then we will have

$$|x_{n+1} - r| \leq \frac{M}{2L}|x_n - r|^2 \quad (\text{E4})$$

Let's denote by ε_1 the error, $|x_1 - r|$, of our initial guess. In fact, let's denote by ε_n the error, $|x_n - r|$, in x_n . Then (E4) says

$$\varepsilon_{n+1} \leq \frac{M}{2L} \varepsilon_n^2$$

In particular

$$\begin{aligned} \varepsilon_2 &\leq \frac{M}{2L} \varepsilon_1^2 \\ \varepsilon_3 &\leq \frac{M}{2L} \varepsilon_2^2 \leq \frac{M}{2L} \left(\frac{M}{2L} \varepsilon_1^2 \right)^2 = \left(\frac{M}{2L} \right)^3 \varepsilon_1^4 \\ \varepsilon_4 &\leq \frac{M}{2L} \varepsilon_3^2 \leq \frac{M}{2L} \left[\left(\frac{M}{2L} \right)^3 \varepsilon_1^4 \right]^2 = \left(\frac{M}{2L} \right)^7 \varepsilon_1^8 \\ \varepsilon_5 &\leq \frac{M}{2L} \varepsilon_4^2 \leq \frac{M}{2L} \left[\left(\frac{M}{2L} \right)^7 \varepsilon_1^8 \right]^2 = \left(\frac{M}{2L} \right)^{15} \varepsilon_1^{16} \end{aligned}$$

By now we can see a pattern forming, that is easily verified by induction⁸.

$$\varepsilon_n \leq \left(\frac{M}{2L} \right)^{2^{n-1}-1} \varepsilon_1^{2^{n-1}} = \frac{2L}{M} \left(\frac{M}{2L} \varepsilon_1 \right)^{2^{n-1}} \quad (\text{E5})$$

As long as $\frac{M}{2L} \varepsilon_1 < 1$ (which gives us a quantitative idea as to how good our first guess has to be in order for Newton's method to work), this goes to zero extremely quickly as n increases. For example, suppose that $\frac{M}{2L} \varepsilon_1 \leq \frac{1}{2}$. Then

$$\varepsilon_n \leq \frac{2L}{M} \left(\frac{1}{2} \right)^{2^{n-1}} \leq \frac{2L}{M} \cdot \begin{cases} 0.25 & \text{if } n = 2 \\ 0.0625 & \text{if } n = 3 \\ 0.0039 = 3.9 \times 10^{-3} & \text{if } n = 4 \\ 0.000015 = 1.5 \times 10^{-5} & \text{if } n = 5 \\ 0.00000000023 = 2.3 \times 10^{-10} & \text{if } n = 6 \\ 0.00000000000000000054 = 5.4 \times 10^{-20} & \text{if } n = 7 \end{cases}$$

Each time you increase n by one, the number of zeroes after the decimal place roughly doubles. You can see why from (E5). Since

$$\left(\frac{M}{2L} \varepsilon_1 \right)^{2^{(n+1)}-1} = \left(\frac{M}{2L} \varepsilon_1 \right)^{2^{n-1} \times 2} = \left[\left(\frac{M}{2L} \varepsilon_1 \right)^{2^{n-1}} \right]^2$$

we have, *very* roughly speaking, $\varepsilon_{n+1} \approx \varepsilon_n^2$. This *quadratic* behaviour is the reason that Newton's method is so useful.

8 Mathematical induction is a technique for proving a sequence $S_1, S_2, S_3, \dots$ of statements. That technique consists of first proving that S_1 is true, and then proving that, for any natural number n , if S_n is true then S_{n+1} is true.

Example C.2.1 (Example C.1.2 continued)

Let's consider, as we did in Example C.1.2, $f(x) = x^2 - 2$, starting with $x_1 = \frac{3}{2}$. Then

$$f'(x) = 2x \quad f''(x) = 2$$

Recalling, from (H1) and (H2), that L is a lower bound on $|f'|$ and M is an upper bound on $|f''|$, we may certainly take $M = 2$ and if, for example, $x_n \geq 1$ for all n (as happened in Example C.1.2), we may take $L = 2$ too. While we do not know what r is, we do know that $1 \leq r \leq 2$ (since $f(1) = 1^2 - 2 < 0$ and $f(2) = 2^2 - 2 > 0$). As we took $x_1 = \frac{3}{2}$, we have $\varepsilon_1 = |x_1 - r| \leq \frac{1}{2}$, so that $\frac{M}{2L}\varepsilon_1 \leq \frac{1}{4}$ and

$$\varepsilon_{n+1} \leq \frac{2L}{M} \left(\frac{M}{2L} \varepsilon_1 \right)^{2^{n-1}} \leq 2 \left(\frac{1}{4} \right)^{2^{n-1}} \quad (\text{E6})$$

This tends to zero very quickly as n increases. Furthermore this is an upper bound on the error and not the actual error. In fact (E6) is a very crude upper bound. For example, setting $n = 3$ gives the bound

$$\varepsilon_4 \leq 2 \left(\frac{1}{4} \right)^{2^2} = 7 \times 10^{-3}$$

and we saw in Example C.1.2 that the actual error in x_4 was smaller than 5×10^{-10} .

Example C.2.1

Example C.2.2 (Example C.1.3 continued)

Let's consider, as we did in Example C.1.3, $f(x) = \sin x$, starting with $x_1 = 3$. Then

$$f'(x) = \cos x \quad f''(x) = -\sin x$$

As $|\sin x| \leq 1$, we may certainly take $M = 1$. In Example C.1.3, all x_n 's were between 3 and 3.2. Since (to three decimal places)

$$\sin(3) = 0.141 > 0 \quad \sin(3.2) = -0.058 < 0$$

the IVT (intermediate value theorem) tells us that $3 < r < 3.2$ and $\varepsilon_1 = |x_1 - r| < 0.2$.

So r and all x_n 's and hence all c 's lie in the interval $(3, 3.2)$. Since

$$-0.9990 = \cos(3) < \cos c < \cos(3.2) = -0.9983$$

we necessarily have $|f'(c)| = |\cos c| \geq 0.9$ and we may take $L = 0.9$. So

$$\varepsilon_{n+1} \leq \frac{2L}{M} \left(\frac{M}{2L} \varepsilon_1 \right)^{2^{n-1}} \leq \frac{2 \times 0.9}{1} \left(\frac{1}{2 \times 0.9} 0.2 \right)^{2^{n-1}} \leq 2 \left(\frac{1}{9} \right)^{2^{n-1}}$$

This tends to zero very quickly as n increases.

Example C.2.2

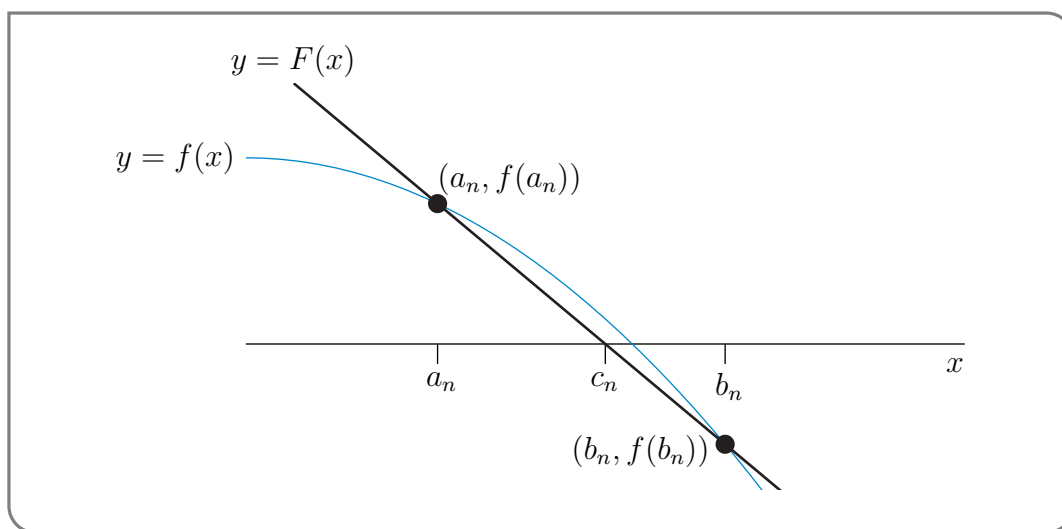
We have now seen two procedures for finding roots of a function $f(x)$ — the bisection method (which does not use the derivative of $f(x)$, but which is not very efficient) and Newton's method (which does use the derivative of $f(x)$, and which is very efficient). In fact, there is a whole constellation of other methods⁹ and the interested reader should search engine their way to, for example, Wikipedia's article on root finding algorithms. Here, we will just mention two other methods, one being a variant of the bisection method and the other being a variant of Newton's method.

C.3 ▲ The false position (regula falsi) method

Let $f(x)$ be a continuous function and let $a_1 < b_1$ with $f(a_1)$ and $f(b_1)$ being of opposite sign.

As we have seen, the bisection method generates a sequence of intervals $I_n = [a_n, b_n]$, $n = 1, 2, 3, \dots$ with, for each n , $f(a_n)$ and $f(b_n)$ having opposite sign (so that, by continuity, f has a root in I_n). Once we have I_n , we choose I_{n+1} based on the sign of f at the midpoint, $\frac{a_n+b_n}{2}$, of I_n . Since we always test the midpoint, the possible error decreases by a factor of 2 each step.

The false position method tries to make the whole procedure more efficient by testing the sign of f at a point that is closer to the end of I_n where the magnitude of f is smaller. To be precise, we approximate $y = f(x)$ by the equation of the straight line through $(a_n, f(a_n))$ and $(b_n, f(b_n))$.



The equation of that straight line is

$$y = F(x) = f(a_n) + \frac{f(b_n) - f(a_n)}{b_n - a_n}(x - a_n)$$

9 What does it say about mathematicians that they have developed so many ways of finding zero?

Then the false position method tests the sign of $f(x)$ at the value of x where $F(x) = 0$.

$$F(x) = f(a_n) + \frac{f(b_n) - f(a_n)}{b_n - a_n}(x - a_n) = 0$$

$$\iff x = a_n - \frac{b_n - a_n}{f(b_n) - f(a_n)}f(a_n) = \frac{a_nf(b_n) - b_nf(a_n)}{f(b_n) - f(a_n)}$$

So once we have the interval I_n , the false position method generates the interval I_{n+1} by the following rule.¹⁰

Equation C.3.1 (false position method).

Set $c_n = \frac{a_nf(b_n) - b_nf(a_n)}{f(b_n) - f(a_n)}$. If $f(c_n)$ has the same sign as $f(a_n)$, then

$$I_{n+1} = [a_{n+1}, b_{n+1}] \quad \text{with} \quad a_{n+1} = c_n, \quad b_{n+1} = b_n$$

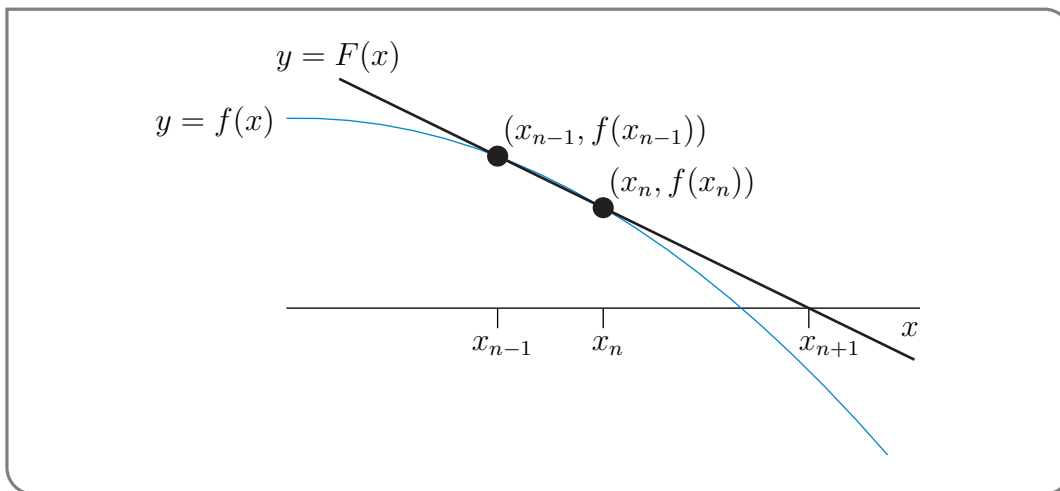
and if $f(c_n)$ and $f(a_n)$ have opposite signs, then

$$I_{n+1} = [a_{n+1}, b_{n+1}] \quad \text{with} \quad a_{n+1} = a_n, \quad b_{n+1} = c_n$$

C.4 ▲ The secant method

Let $f(x)$ be a continuous function. The secant method is a variant of Newton's method that avoids the use of the derivative of $f(x)$ — which can be very helpful when dealing with the derivative is not easy. It avoids the use of the derivative by approximating $f'(x)$ by $\frac{f(x+h)-f(x)}{h}$ for some h . That is, it approximates the tangent line to f at x by a secant line to f that passes through x . To limit the number of evaluations of $f(x)$ required, it uses $x = x_{n-1}$ and $x + h = x_n$. Here is how it works.

Suppose that we have already found x_n . Then we denote by $y = F(x)$ the equation of the (secant) line that passes through $(x_{n-1}, f(x_{n-1}))$ and $(x_n, f(x_n))$ and we choose x_{n+1} to be the value of x where $F(x) = 0$.



10 The convergence behaviour of the false position method is relatively complicated. So we do not discuss it here. As always, we invite the interested reader to visit their favourite search engine.

The equation of the secant line is

$$y = F(x) = f(x_{n-1}) + \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}(x - x_{n-1})$$

so that x_{n+1} is determined by

$$\begin{aligned} 0 = F(x_{n+1}) &= f(x_{n-1}) + \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}(x_{n+1} - x_{n-1}) \\ \iff x_{n+1} &= x_{n-1} - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}f(x_{n-1}) \end{aligned}$$

or, simplifying,

Equation C.4.1 (secant method).

$$x_{n+1} = \frac{x_{n-1}f(x_n) - x_nf(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Of course, to get started with $n = 1$, we need two initial guesses, x_0 and x_1 , for the root.

Example C.4.2 (Approximating $\sqrt{2}$, again)

In this example we compute, approximately, the square root of two by applying the secant method to the equation

$$f(x) = x^2 - 2 = 0$$

and we'll compare the secant method results with the corresponding Newton's method results. (See Example [C.1.2](#).)

Since $f'(x) = 2x$, [\(C.1.1\)](#) says that, under Newton's method, we should iteratively apply

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n}$$

while [\(C.4.1\)](#) says that, under the secant method, we should iteratively apply (after a little simplifying algebra)

$$\begin{aligned} x_{n+1} &= \frac{x_{n-1}f(x_n) - x_nf(x_{n-1})}{f(x_n) - f(x_{n-1})} = \frac{x_{n-1}[x_n^2 - 2] - x_n[x_{n-1}^2 - 2]}{x_n^2 - x_{n-1}^2} \\ &= \frac{x_{n-1}x_n[x_n - x_{n-1}] + 2[x_n - x_{n-1}]}{x_n^2 - x_{n-1}^2} \\ &= \frac{x_{n-1}x_n + 2}{x_{n-1} + x_n} \end{aligned}$$

Here are the results, starting Newton's method with $x_1 = 4$ and starting the secant method

with $x_0 = 4$, $x_1 = 3$. (So we are giving the secant method a bit of a head start.)

	secant method	Newton's method
x_0	4	
x_1	3	4
x_2	2	2.25
x_3	1.6	1.57
x_4	1.444	1.422
x_5	1.4161	1.414234
x_6	1.414233	1.414213562525
x_7	1.414213575	1.414213562373095

For comparison purposes, the square root of 2, to 15 decimal places, is 1.414213562373095. So the secant method x_7 is accurate to 7 decimal places and the Newton's method x_7 is accurate to at least 15 decimal places.

Example C.4.2

The advantage that the secant method has over Newton's method is that it does not use the derivative of f . This can be a substantial advantage, for example when evaluation of the derivative is computationally difficult or expensive. On the other hand, the above example suggests that the secant method is not as fast as Newton's method. The following section shows that this is indeed the case.

C.5 ▲ The Error Behaviour of the Secant Method

Let $f(x)$ have two continuous derivatives, and let r be any solution of $f(x) = 0$. We will now get a pretty good handle on the error behaviour of the secant method near r .

Denote by $\tilde{\epsilon}_n = x_n - r$ the (signed) error in x_n and by $\epsilon_n = |x_n - r|$ the (absolute) error in x_n . Then, $x_n = r + \tilde{\epsilon}_n$, and, by (C.4.1),

$$\begin{aligned}\tilde{\epsilon}_{n+1} &= \frac{x_{n-1}f(x_n) - x_nf(x_{n-1})}{f(x_n) - f(x_{n-1})} - r \\ &= \frac{[r + \tilde{\epsilon}_{n-1}]f(x_n) - [r + \tilde{\epsilon}_n]f(x_{n-1})}{f(x_n) - f(x_{n-1})} - r \\ &= \frac{\tilde{\epsilon}_{n-1}f(x_n) - \tilde{\epsilon}_nf(x_{n-1})}{f(x_n) - f(x_{n-1})}\end{aligned}$$

By the Taylor expansion (3.4.32) and the mean value theorem (Theorem 2.13.4),

$$\begin{aligned}f(x_n) &= f(r) + f'(r)\tilde{\epsilon}_n + \frac{1}{2}f''(c_1)\tilde{\epsilon}_n^2 \\ &= f'(r)\tilde{\epsilon}_n + \frac{1}{2}f''(c_1)\tilde{\epsilon}_n^2 \\ f(x_n) - f(x_{n-1}) &= f'(c_2)[x_n - x_{n-1}] \\ &= f'(c_2)[\tilde{\epsilon}_n - \tilde{\epsilon}_{n-1}]\end{aligned}$$

for some c_1 between r and x_n and some c_2 between x_{n-1} and x_n . So, for x_{n-1} and x_n near r , c_1 and c_2 also have to be near r and

$$\begin{aligned} f(x_n) &\approx f'(r)\tilde{\varepsilon}_n + \frac{1}{2}f''(r)\tilde{\varepsilon}_n^2 \\ f(x_{n-1}) &\approx f'(r)\tilde{\varepsilon}_{n-1} + \frac{1}{2}f''(r)\tilde{\varepsilon}_{n-1}^2 \\ f(x_n) - f(x_{n-1}) &\approx f'(r)[\tilde{\varepsilon}_n - \tilde{\varepsilon}_{n-1}] \end{aligned}$$

and

$$\begin{aligned} \tilde{\varepsilon}_{n+1} &= \frac{\tilde{\varepsilon}_{n-1}f(x_n) - \tilde{\varepsilon}_nf(x_{n-1})}{f(x_n) - f(x_{n-1})} \\ &\approx \frac{\tilde{\varepsilon}_{n-1}[f'(r)\tilde{\varepsilon}_n + \frac{1}{2}f''(r)\tilde{\varepsilon}_n^2] - \tilde{\varepsilon}_n[f'(r)\tilde{\varepsilon}_{n-1} + \frac{1}{2}f''(r)\tilde{\varepsilon}_{n-1}^2]}{f'(r)[\tilde{\varepsilon}_n - \tilde{\varepsilon}_{n-1}]} \\ &= \frac{\frac{1}{2}\tilde{\varepsilon}_{n-1}\tilde{\varepsilon}_nf''(r)[\tilde{\varepsilon}_n - \tilde{\varepsilon}_{n-1}]}{f'(r)[\tilde{\varepsilon}_n - \tilde{\varepsilon}_{n-1}]} \\ &= \frac{f''(r)}{2f'(r)}\tilde{\varepsilon}_{n-1}\tilde{\varepsilon}_n \end{aligned}$$

Taking absolute values, we have

$$\varepsilon_{n+1} \approx K\varepsilon_{n-1}\varepsilon_n \quad \text{with } K = \left| \frac{f''(r)}{2f'(r)} \right| \quad (\text{E7})$$

We have seen that Newton's method obeys a similar formula — (E3) says that, when x_n is near r , Newton's method obeys $\varepsilon_{n+1} \approx K\varepsilon_n^2$, also with $K = \left| \frac{f''(r)}{2f'(r)} \right|$. As we shall now see, the change from ε_n^2 in $\varepsilon_{n+1} \approx K\varepsilon_n^2$, to $\varepsilon_{n-1}\varepsilon_n$ in $\varepsilon_{n+1} \approx K\varepsilon_{n-1}\varepsilon_n$, does have a substantial impact on the behaviour of ε_n for large n .

To see the large n behaviour, we now iterate (E7). The formulae will look simpler if we multiply (E7) by K and write $\delta_n = K\varepsilon_n$. Then (E7) becomes $\delta_{n+1} \approx \delta_{n-1}\delta_n$ (and we have eliminated K). The first iterations are

$$\begin{aligned} \delta_2 &\approx \delta_0\delta_1 \\ \delta_3 &\approx \delta_1\delta_2 \approx \delta_0\delta_1^2 \\ \delta_4 &\approx \delta_2\delta_3 \approx \delta_0^2\delta_1^3 \\ \delta_5 &\approx \delta_3\delta_4 \approx \delta_0^3\delta_1^5 \\ \delta_6 &\approx \delta_4\delta_5 \approx \delta_0^5\delta_1^8 \\ \delta_7 &\approx \delta_5\delta_6 \approx \delta_0^8\delta_1^{13} \end{aligned}$$

Notice that every δ_n is of the form $\delta_0^{\alpha_n}\delta_1^{\beta_n}$. Substituting $\delta_n = \delta_0^{\alpha_n}\delta_1^{\beta_n}$ into $\delta_{n+1} \approx \delta_{n-1}\delta_n$ gives

$$\delta_0^{\alpha_{n+1}}\delta_1^{\beta_{n+1}} \approx \delta_0^{\alpha_{n-1}}\delta_1^{\beta_{n-1}}\delta_0^{\alpha_n}\delta_1^{\beta_n}$$

and we have

$$\alpha_{n+1} = \alpha_{n-1} + \alpha_n \quad \beta_{n+1} = \beta_{n-1} + \beta_n \quad (\text{E8})$$

The recursion rule in (E8) is famous¹¹. The Fibonacci¹² sequence (which is 0, 1, 1, 2, 3, 5, 8, 13, $\dots$), is defined by

$$\begin{aligned} F_0 &= 0 \\ F_1 &= 1 \\ F_n &= F_{n-1} + F_{n-2} \quad \text{for } n > 1 \end{aligned}$$

So, for $n \geq 2$, $\alpha_n = F_{n-1}$ and $\beta_n = F_n$ and

$$\delta_n \approx \delta_0^{\alpha_n} \delta_1^{\beta_n} = \delta_0^{F_{n-1}} \delta_1^{F_n}$$

One of the known properties of the Fibonacci sequence is that, for large n ,

$$F_n \approx \frac{\varphi^n}{\sqrt{5}} \quad \text{where } \varphi = \frac{1 + \sqrt{5}}{2} \approx 1.61803$$

This φ is the golden ratio¹³. So, for large n ,

$$\begin{aligned} K\varepsilon_n = \delta_n &\approx \delta_0^{F_{n-1}} \delta_1^{F_n} \approx \delta_0^{\frac{\varphi^{n-1}}{\sqrt{5}}} \delta_1^{\frac{\varphi^n}{\sqrt{5}}} = \delta_0^{\frac{1}{\sqrt{5}\varphi} \times \varphi^n} \delta_1^{\frac{1}{\sqrt{5}} \times \varphi^n} \\ &= d^{\varphi^n} \quad \text{where } d = \delta_0^{\frac{1}{\sqrt{5}\varphi}} \delta_1^{\frac{1}{\sqrt{5}}} \\ &\approx d^{1.6^n} \end{aligned}$$

Assuming that $0 < \delta_0 = K\varepsilon_0 < 1$ and $0 < \delta_1 = K\varepsilon_1 < 1$, we will have $0 < d < 1$.

By way of contrast, for Newton's method, for large n ,

$$K\varepsilon_n \approx d^{2^n} \quad \text{where } d = (K\varepsilon_1)^{1/2}$$

As 2^n grows quite a bit more quickly than 1.6^n (for example, when $n=5$, $2^n = 32$ and $1.6^n = 10.5$, and when $n = 10$, $2^n = 1024$ and $1.6^n = 110$) Newton's method homes in on the root quite a bit faster than the secant method, assuming that you start reasonably close to the root.

11 Plug "Fibonacci sequence in nature" into your search engine of choice.

12 Fibonacci (1170-1250) was an Italian mathematician who was also known as Leonardo of Pisa, Leonardo Bonacci and Leonardo Biglio Pisano.

13 Also worth a quick trip to your search engine.